

Cours 9 – Apprentissage de préférences, élicitation incrémentale.

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Preference data

- regression examples : target values $f_w(x) = y$
- pairwise comparisons : preference constraints $f_w(x) \geq f_w(y)$
- batch/online
- may be costly to obtain (Interactions with the DM)

Learning f_w from preference data : main challenges

- parameters are subject to consistency constraints (monotonicity, risk-aversion, fairness),
- high number of parameters, small databases of preferences examples : need for sparse representations,
- interactions between parameters (weights and utilities).
- preference learning on implicit sets (comb. or cont. domains).

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Focus on decision models linear in their parameter

For any given x, y

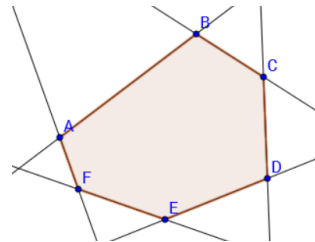
$$f_{\alpha w}(x) = \alpha f_w(x) \text{ for all } \alpha > 0 \quad \text{and} \quad f_{v+w}(x) = f_v(x) + f_w(x)$$

- Regression examples yield linear equations (in w) :

$$f_w(x_1, \dots, x_n) = y$$

- Databases of preference examples ($x \succeq y$) yield systems of linear inequalities :

$$f_w(x_1, \dots, x_n) \geq f_w(y_1, \dots, y_n)$$



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Focus on the discrete Choquet integral

$N = \{1, \dots, n\}$ criteria, $v : 2^N \rightarrow [0, 1]$ “capacity” or “fuzzy measure”

$$v(\emptyset) = 0, v(N) = 1, v(A) \leq v(B) \text{ if } A \subseteq B \text{ (monotonicity)}$$

CIU : Choquet Integral of Utilities [Schmeidler, 1989; Grabisch, 1996]

$$C_v^u(x) = C_v(u_1(x_1), \dots, u_n(x_n)) = \sum_{i=1}^n [v(X_{(i)}) - v(X_{(i-1)})] u_{(i)}(x_{(i)})$$

If $\alpha \leq \beta \leq \gamma$

$$\begin{aligned} C_v(\alpha, \beta, \gamma) &= \alpha v(\{1, 2, 3\}) + (\beta - \alpha) v(\{2, 3\}) + (\gamma - \beta) v(\{3\}) \\ &= \alpha [v(\{1, 2, 3\}) - v(\{2, 3\})] + \beta [v(\{2, 3\}) - v(\{3\})] + \gamma v(\{3\}) \end{aligned}$$

- if v additive ($v(A \cup B) + v(A \cap B) = v(A) + v(B)$)
 C_v^u = weighted sum of utilities
- if v non-additive : $v(A \cup B) \neq v(A) + v(B)$ for some $A, B : A \cap B = \emptyset$, model allowing positive or negative synergies (criteria interactions)
- if v convex $v(A \cup B) + v(A \cap B) \geq v(A) + v(B)$ and u_i concave : C_v^u concave model, favoring balanced utility vectors (fairness, risk aversion)

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Learning CIU : Motivations and Challenges

Motivations

- $C_v^u(x)$ is linear in v
- allows to model interactions among criteria
- includes many simpler decision models as special cases
- admits different extensions that can be learned similarly

Challenges

- uses a complex weighting parameter (2^n coefficients) subject to monotonicity constraints
- combines two parameters u and v interacting in the model

1. Learning utility functions u_i

$$C_v^u(x) = C_v(u_1(x_1), \dots, u_n(x_n)) = \sum_{i=1}^n [v(X_{(i)}) - v(X_{(i-1)})] u_{(i)}(x_{(i)})$$

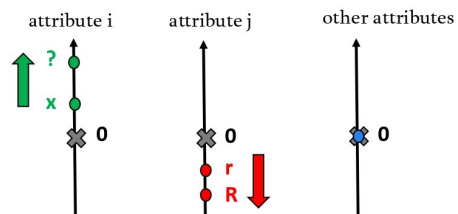
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Learning the Utility Functions

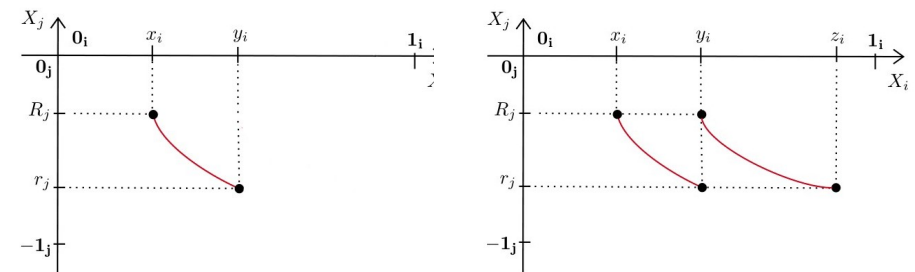
Knowledge extraction on the utility functions with properly selected indifference queries

Adaptation and extension of a method from Decision-Making Under Risk [Wakker and Deneffe, 1996 ; Abdellaoui 2000]



Restricted Solvability Assumption [krantz et al., 1971]

Tradeoff Queries



Equality constraint on u_i , independent of v

$$u_i(y_i) - u_i(x_i) = u_i(z_i) - u_i(y_i)$$

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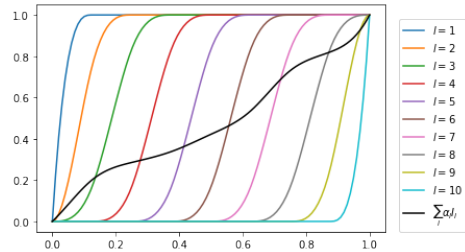
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Monotonic Spline Regression

1) Preference data : $(x_i^k, y_i^k, z_i^k)_{k=1}^K$ obtained for various $j \in N$ and $R_j, r_j \in X_j$ and starting points x_i such that :

$$u_i(y_i^k) - u(z_i^k) = u_i(y_i^k) - u(x_i^k), \quad k = 1, \dots, K$$

2) Constraint-based l-spline regression : $u_i(x) = \sum_l \alpha_l l_i(x)$



$$\begin{aligned} & \min \sum_k |\epsilon_k| \\ \text{s.t. } & \epsilon_k = (u_i(y^k) - u_i(x^k)) - (u_i(y^k) - u_i(z^k)), \quad k = 1, \dots, K \end{aligned}$$

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Example

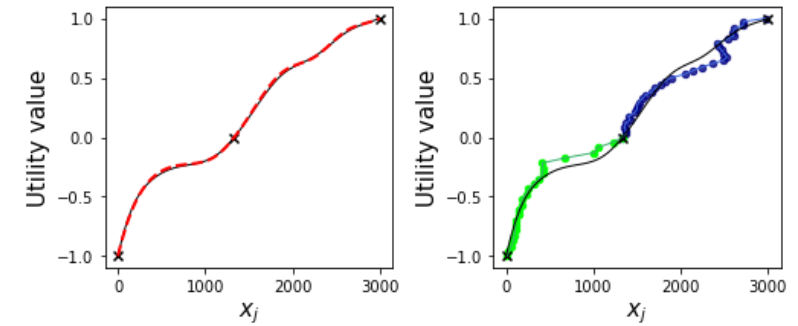


Figure – Ground truth utility function (black) and learned utility function with the monotonic constraint-based regression (left ; red) and with standard sequences (right ; blue and green points)

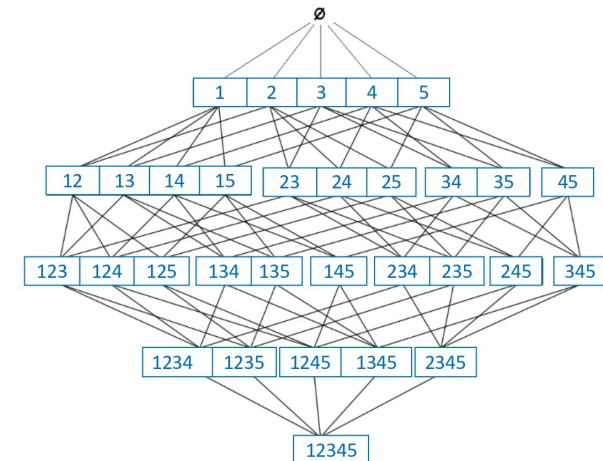
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2. Learning the capacity

$$C_v^u(x) = C_v(u_1(x_1), \dots, u_n(x_n)) = \sum_{i=1}^n [v(X_{(i)}) - v(X_{(i-1)})] u_{(i)}(x_{(i)})$$

The inclusion lattice of a capacity

Example with $N = \{1, 2, 3, 4, 5\}$



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Möbius representation of capacities and associated models

Möbius transform

$$\forall S \subseteq N, \quad m_v(S) = \sum_{T \subseteq S} (-1)^{|S \setminus T|} v(T) \quad \text{and} \quad v(S) = \sum_{T \subseteq S} m_v(T)$$

Choquet Integral

$$\begin{aligned} C_v(x) &= \sum_{S \subseteq N} m_v(S) \min_{i \in S} \{x_i\} && \text{conjunctive form} \\ C_v(x) &= \sum_{S \subseteq N} m_{\bar{v}}(S) \max_{i \in S} \{x_i\} && \text{disjunctive form} \end{aligned}$$

with $\bar{v} : S \rightarrow v(N) - v(N \setminus S)$ the dual of v .

Multilinear model

$$ML_v(x) = \sum_{S \subseteq N} m_v(S) \prod_{i \in S} x_i$$

General model

$$F(x) = \sum_{S \subseteq N} m_S \phi_S(x_S)$$

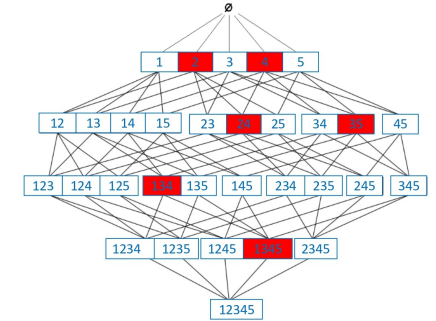
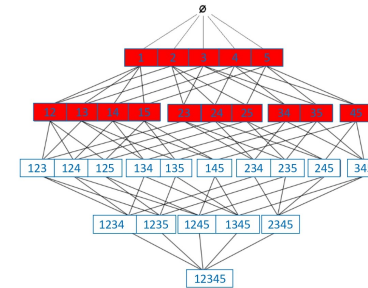
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Compact Möbius representations of capacities

$$m_v(\{i, j\}) = v(\{i, j\}) - [v(\{i\}) + v(\{j\})] \text{ gap to additivity}$$

• Standard approach : k-additivity

• Selecting the relevant coalitions



Descriptive limit : simple n-additive models

- $f(x) = \min_i \{x_i\}$ (egalitarianism)
- $f_\alpha(x) = \alpha \min_i \{x_i\} + (1 - \alpha) \max_i \{x_i\}$ (Hurwicz).

Non-null Möbius masses

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Sparse Möbius representation of Choquet Integrals

Proposition

For any monotonic capacity v , $\|\mathbf{m}_v\|_0 \leq \|\mathbf{v}\|_0$
($\|\cdot\|_0$: number of non-null coefficients)

Example

Let $N = \{1, 2, 3\}$ and $v, \bar{v}$ defined on N by :

S	1	2	3	1, 2	1, 3	2, 3	1, 2, 3
$v(S)$	0.1	0.2	0.3	0.3	0.4	0.5	1.0
$m_v(S)$	0.1	0.2	0.3	0.0	0.0	0.0	0.4

$$C_v(x) = 0.1x_1 + 0.2x_2 + 0.3x_3 + 0.4 \min\{x_1, x_2, x_3\}$$

S	1	2	3	1, 2	1, 3	2, 3	1, 2, 3
$v(S)$	0.5	0.6	0.7	0.7	0.8	0.9	1.0
$m_v(S)$	0.5	0.6	0.7	-0.4	-0.4	-0.4	0.4
$m_{\bar{v}}(S)$	0.1	0.2	0.3	0.0	0.0	0.0	0.4

$$C_v(x) = 0.1x_1 + 0.2x_2 + 0.3x_3 + 0.4 \max\{x_1, x_2, x_3\}$$

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Learning a Sparse Möbius Representation of the Capacity

- utilities are known ! $x_i := u_i(x_i)$
- $\phi(x) = (\phi_S(x))_{S \subseteq N} = (\min_{i \in S} \{x_i\})_{S \subseteq N}$
- $\mathbf{m} = (m_v(S))_{S \subseteq N}$

Choquet and extensions seen as a linear model

$$F(x) = \langle \mathbf{m}, \phi(\mathbf{x}) \rangle$$

- Preference statements $\{(x^\ell, y^\ell) \in \mathcal{X}^2 : x^\ell \succ y^\ell, \ell \in P\}$
- Indifference statements $\{(x^\ell, y^\ell) \in \mathcal{X}^2 : x^\ell \sim y^\ell, \ell \in I\}$
- $\lambda > 0$: regularization hyper-parameter ;

L_1 -norm regularization on the möbius representation (LAD-LASSO)

$$\min \sum_{\ell \in I} |\langle \mathbf{m}, \phi(x^\ell) \rangle - \langle \mathbf{m}, \phi(y^\ell) \rangle| + \sum_{\ell \in P} (\langle \mathbf{m}, \phi(x^\ell) \rangle - \langle \mathbf{m}, \phi(y^\ell) \rangle)^- + \lambda \sum_{S \subseteq N, |S| > 1} |m_v(S)|$$

$$\sum_{B \subseteq S, B \ni i} m_v(B) \geq 0, \quad \forall S \subseteq N, \forall i \in S \quad (1)$$

$$\sum_{S \subseteq N} m_v(S) = 1 \quad (2)$$

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Adaptive LASSO

Adaptive L_1 -penalty (ADA- L_1)

Problem $\mathcal{P}'$

$$\min \sum_{i \in I} |C_v^u(x^i) - C_v^u(y^i)| + \sum_{i \in P} (C_v^u(x^i) - C_v^u(y^i))^- + \sum_{S \subseteq N, |S| > 1} \lambda_{\rho(S)} |m_v(S)|$$

s.t. 1, 2

where $\lambda_{\rho(S)} = \lambda / |\hat{m}_v(S)|$ for all $S \subseteq N$ and $\hat{m}_v$ is the optimal solution of the following initialization problem :

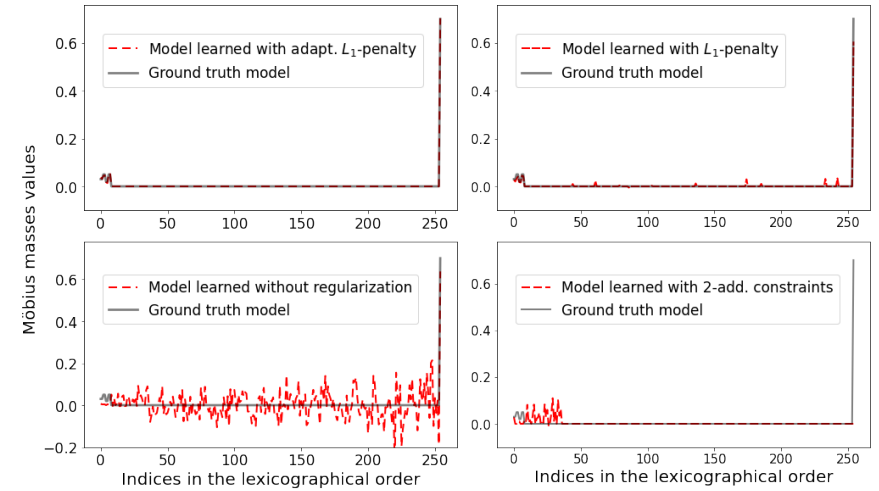
Problem $\mathcal{P}_0'$

$$\min \sum_{i \in I} |C_v^u(x^i) - C_v^u(y^i)| + \sum_{i \in P} (C_v^u(x^i) - C_v^u(y^i))^- + \lambda_2 \|m_v\|_2^2$$

s.t. 1, 2

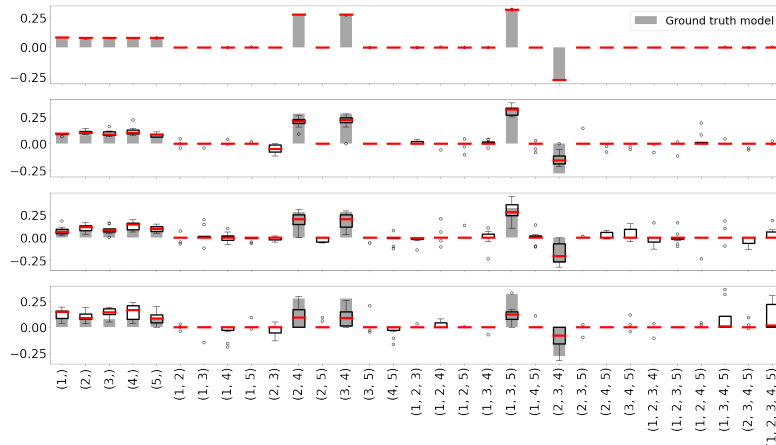
Example 1 : the augmented minimum

$$F(x) = (1 - \epsilon) \min\{u_i(x_i), i \in N\} + \frac{\epsilon}{n} \sum_{i=1}^n u_i(x_i)$$



Learned models and hidden model (min model)

Example 2 : testing sensitivity to noise



Ground truth model (grey bar) and boxplots of the learned Möbius masses over 10 simulations for an increasing noise amplitude $\sigma \in \{0, 0.03, 0.05, 0.1\}$ from top to bottom.

Scalability

Dealing with an exponential number of constraints

Monotonicity constraints

$$\sum_{B \subseteq S, B \ni i} m_v(B) \geq 0, \quad \forall S \subseteq N, \forall i \in S$$

- Exponential number of constraints $C(n) = \sum_{k=1}^n k \binom{n}{k}$.
- Constraint generation** : in practice a small fraction $\tilde{C}(n)$ of the monotonicity constraints are needed to reach the optimal solution.

n	$\tilde{C}(n)$	$C(n)$	Time ESG	Time ESC
6	3.2 ± 6.4	192	0.6 ± 0.2	0.6 ± 0.1
9	2.4 ± 7.2	2304	4.2 ± 1.9	18.0 ± 4.6
12	151.9 ± 222.2	24576	61.0 ± 30.4	1212.6 ± 247.6
15	2777.6 ± 4326.5	245760	3448.6 ± 5613.1	-

ESG : Exact solving with constraint generation

ESC : Exact solving fully constrained.

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Dealing with an exponential number of variables

$$(\mathcal{P}) \min \sum_{\ell \in P} \epsilon_\ell + \sum_{\ell \in I} (\epsilon_\ell^- + \epsilon_\ell^+) + \lambda \sum_{j=n+1}^{2^n} |m_j|$$

$$\langle \mathbf{m}, \phi(\mathbf{x}^\ell) \rangle - \langle \mathbf{m}, \phi(\mathbf{y}^\ell) \rangle + \epsilon_\ell \geq \delta, \quad \ell \in P \quad (3)$$

$$\langle \mathbf{m}, \phi(\mathbf{x}^\ell) \rangle - \langle \mathbf{m}, \phi(\mathbf{y}^\ell) \rangle + \epsilon_\ell^+ - \epsilon_\ell^- = 0, \quad \ell \in I \quad (4)$$

$$\langle \mathbf{m}, \mathbf{1} \rangle = 1 \quad (5)$$

$$\epsilon_\ell \geq 0, \quad \ell \in P, \quad \epsilon_\ell^+, \epsilon_\ell^- \geq 0, \quad \ell \in I \quad (6)$$

IRLS Algorithm

Consider the IRLS sequence $\mathbf{m}^{(k)}$ initialized with $\mathbf{m}^{(0)} = \mathbf{1}$ such that :

$$\mathbf{m}^{(k+1)} \in \operatorname{argmin} \sum_{\ell \in P} \epsilon_\ell + \sum_{\ell \in I} (\epsilon_\ell^- + \epsilon_\ell^+) + \sum_{j>n} \lambda_j^{(k)} m_j^2 \quad (\mathcal{P}_k)$$

$$\text{s.t. (3), (4), (5), (6)} \quad \lambda_j^{(k)} = (m_j^{(k)^2} + \eta^2)^{-\frac{1}{2}}$$

Convergence (based on [Beck, 2015])

The sequence $\mathbf{m}^{(k+1)}$ converges towards a near-optimal solution of $\mathcal{P}$.

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Efficient dual formulation for $|P| + |I| \ll 2^n$

Interest

$\mathcal{P}_k$ admits a dual formulation $\mathcal{D}_k$ which has $|P| + |I| + 1$ variables and $2(|P| + |I|)$ constraints (**kernel trick**).

Toy example ($|P| = 3, n = 5$)

$$(\mathcal{P}_k) \min_{\mathbf{m} \in \mathbb{R}^{32}} \sum_{i=1}^3 \epsilon_i + \sum_{j=5}^{32} \lambda_j^{(k)} m_j^2$$

$$\langle \mathbf{m}, \phi(\mathbf{x}^1) \rangle - \langle \mathbf{m}, \phi(\mathbf{y}^1) \rangle + \epsilon_1 \geq \delta$$

$$\langle \mathbf{m}, \phi(\mathbf{x}^2) \rangle - \langle \mathbf{m}, \phi(\mathbf{y}^2) \rangle + \epsilon_2 \geq \delta$$

$$\langle \mathbf{m}, \phi(\mathbf{x}^3) \rangle - \langle \mathbf{m}, \phi(\mathbf{y}^3) \rangle + \epsilon_3 \geq \delta$$

$$\epsilon_1, \epsilon_2, \epsilon_3 \geq 0$$

$$(\mathcal{D}_k) \max_{\alpha \in [0,1]^3} \sum_{i,j=1}^3 \alpha_i \alpha_j \delta^{i\top} \mathbf{D}_k^{-1} \delta^j$$

with $\delta^i = \phi(\mathbf{x}^i) - \phi(\mathbf{y}^i)$

and $\mathbf{D}_k = \operatorname{diag}((\lambda_j^{(k)})_{j=n}^{2^n})$

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Preference Kernel

First iteration :

$$\delta^{i\top} \mathbf{D}_k^{-1} \delta^j = \delta^{i\top} \delta^j$$

$$= \langle \phi(\mathbf{x}^i), \phi(\mathbf{x}^j) \rangle + \langle \phi(\mathbf{y}^i), \phi(\mathbf{y}^j) \rangle - \langle \phi(\mathbf{x}^i), \phi(\mathbf{y}^j) \rangle - \langle \phi(\mathbf{y}^i), \phi(\mathbf{x}^j) \rangle$$

Choquet kernel [Tehrani, 2014]

$$\langle \phi(\mathbf{x}), \phi(\mathbf{x}') \rangle = \langle \mathbf{x}, \mathbf{x}' \rangle + \sum_{i=1}^{n-1} x_{(i)} \left\{ \sum_{j=1}^{n-i} 2^{n-i-j} \cdot \min \{x'_{(i)}, x'_{[i+1]_i}\} \right\}$$

where $(.)$ is a permutation of N such that $x_{(i)} \leq x_{(i+1)}$ and $[.]_i$ are permutations sorting each vector $(x'_{(i+1)}, \dots, x'_{(n)})$ by increasing order.

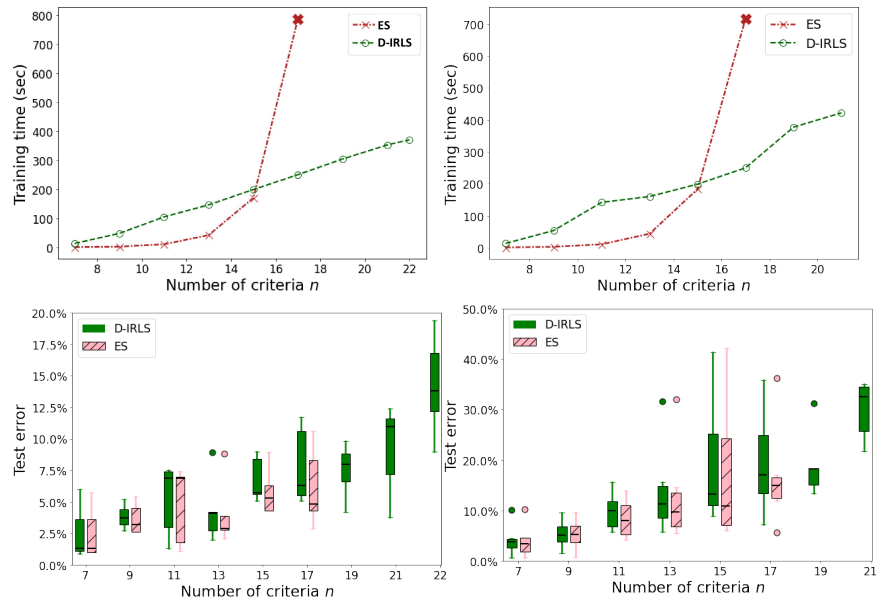
Multilinear kernel

$$\langle \phi(\mathbf{x}), \phi(\mathbf{x}') \rangle = \sum_{S \subseteq N} \prod_{i \in S} x_i \prod_{i \in S} x'_i = \prod_{i=1}^n (x_i x'_i + 1) \quad (7)$$

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Numerical tests on synthetic data

Choquet Integral (left) and Multilinear model (right) ($|P| + |I| = 500$, 10 simulations)



Mean training time and test error boxplot for D-IRLS and ES with multilinear (top) and Choquet Integral (bottom) model.

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Hybrid models : $F(x) = \sum_{S \subseteq N} m_S^1 \phi_S^1(x_S) + m_S^2 \phi_S^2(x_S)$

Hybrid form of the Choquet Integral

$$v = v^\wedge + v^\vee \Rightarrow C_v(x) = C_{v^\wedge}(x) + C_{v^\vee}(x) \\ \Rightarrow C_v(x) = \sum_{S \subseteq N} (m_{v^\wedge}(S) \min_{i \in S} \{x_i\} + m_{v^\vee}(S) \max_{i \in S} \{x_i\})$$

Recovering the Hurwicz model ($F(x) = \frac{1}{2}(\min_{i \in N} \{x_i\} + \max_{i \in N} \{x_i\})$)

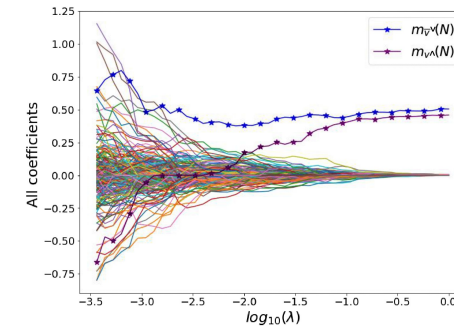
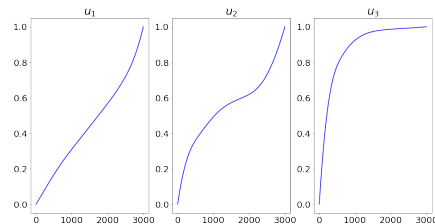


Figure – Selection path for the learning of the Hurwicz model.

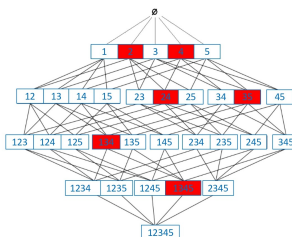
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Summary : A Two-Step Learning Procedure

1- Learning the utility functions using selected preference queries



2- Once the utility functions are known, learning a sparse representation of the capacity using general preference examples



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Active Learning for Decision Making

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Active Learning for Decision Making

Preference learning to make a decision on a given set of alternatives.

Observation

Only a fraction of preference information is necessary to identify the optimal choice within a given set of alternatives.

→ focus the elicitation burden on the useful part of preference information

Active Learning/Incremental Elicitation

ISMAUT [White et al., 1984], Utilities [Wang and Boutilier 2003], Choquet capacities [Benabbou et al, 2017]

- active selection of informative preference queries (Regret minimization)
- progressive reduction of the parameter space defining the capacity
- iteration until a necessary winner is obtained

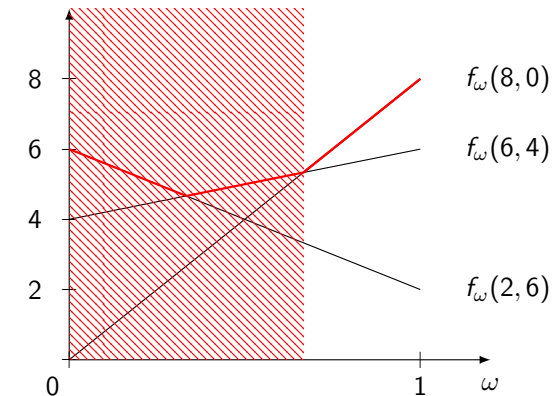
Extensions to deal with uncertainty in preferences data

- Bayesian Linear Regression approach : [Bourdache et al., 2019]
- Possibilistic approach : [Adam and Destercke, 2021]

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Incremental approach : an example

- Alternatives : $\mathcal{X} = \{(2, 6), (6, 4), (8, 0)\}$
- A simple linear model : $f_{\omega}(x) = \omega x_1 + (1 - \omega)x_2 = \omega(x_1 - x_2) + x_2$



All the alternatives are potentially optimal ! "(8, 0) is preferred to (6, 4)"

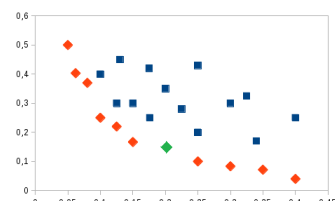
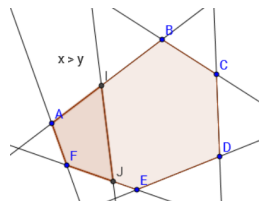
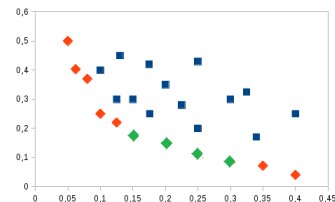
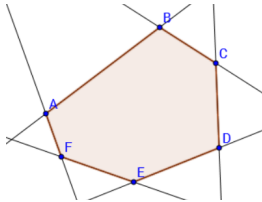
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Exploiting linearities w.r.t the parameter

Key Remark : $f_{\omega}(x) \geq f_{\omega}(y)$ is a linear constraint w.r.t. ω , for any fixed x, y

Ω : Space of weights ω_i

Space of criteria and possible winners



Principle : Iteratively reduce the convex polyhedron until the determination of a necessary winner

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Active learning based on the Minimax Regret criterion

- $\mathcal{P}$: Set of preference statements.
- $\Omega_{\mathcal{P}}$: Set of parameters compatible with $\mathcal{P}$.

PAIRWISE MAX REGRET (PMR) when f_{ω} is to be maximized

$$\text{PMR}(x, y; \Omega_{\mathcal{P}}) = \max_{\omega \in \Omega_{\mathcal{P}}} \{f_{\omega}(y) - f_{\omega}(x)\} \quad \text{LP}$$

where f_{ω} is the aggregator corresponding to parameter ω .

MAX REGRET (MR)

$$\text{MR}(x; \Omega_{\mathcal{P}}) = \max_{y \in \mathcal{X}} \text{PMR}(x, y; \Omega_{\mathcal{P}})$$

MINIMAX REGRET (MMR)

$$\text{MMR}(\Omega_{\mathcal{P}}) = \min_{x \in \mathcal{X}} \text{MR}(x; \Omega_{\mathcal{P}})$$

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Using regrets for possibly optimal solutions

PROPOSITION :

$\forall x \in \mathcal{X}, \text{MR}(x) \leq 0 \Rightarrow x$ is a necessarily optimal solution.

SOME NATURAL PROPERTIES

For any $\mathcal{P}' \supset \mathcal{P}$ we have $\Omega_{\mathcal{P}'} \subseteq \Omega_{\mathcal{P}}$, therefore :

- $\text{PMR}(x, y; \Omega_{\mathcal{P}'}) \leq \text{PMR}(x, y; \Omega_{\mathcal{P}})$ for all $x, y \in \mathcal{X}$
- $\text{MR}(x; \Omega_{\mathcal{P}'}) \leq \text{MR}(x; \Omega_{\mathcal{P}})$ for all $x \in \mathcal{X}$
- $\text{MMR}(\Omega_{\mathcal{P}'}) \leq \text{MMR}(\Omega_{\mathcal{P}})$

→ Asking queries to the user makes MMR decrease.

Two main options

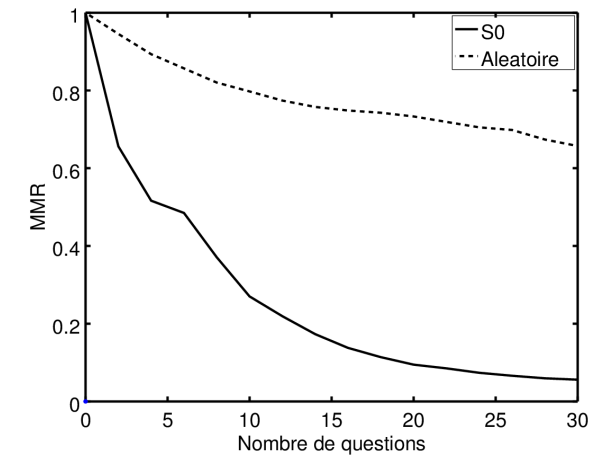
- keep asking preference queries, as informative as possible, until $\text{MMR} \leq 0$, which gives a necessarily optimal solution.
- stop preference queries as soon as $\text{MMR} \leq \lambda$ for some $\lambda > 0$ and recommend the most robust choice :

$$x^* \in \arg \min_{x \in \mathcal{X}} \text{MR}(x; \Omega_{\mathcal{P}})$$

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Example of active learning session

10 criteria and 1000 alternatives, averages over 100 runs



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Some references

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